MŰEGYETEM 1782

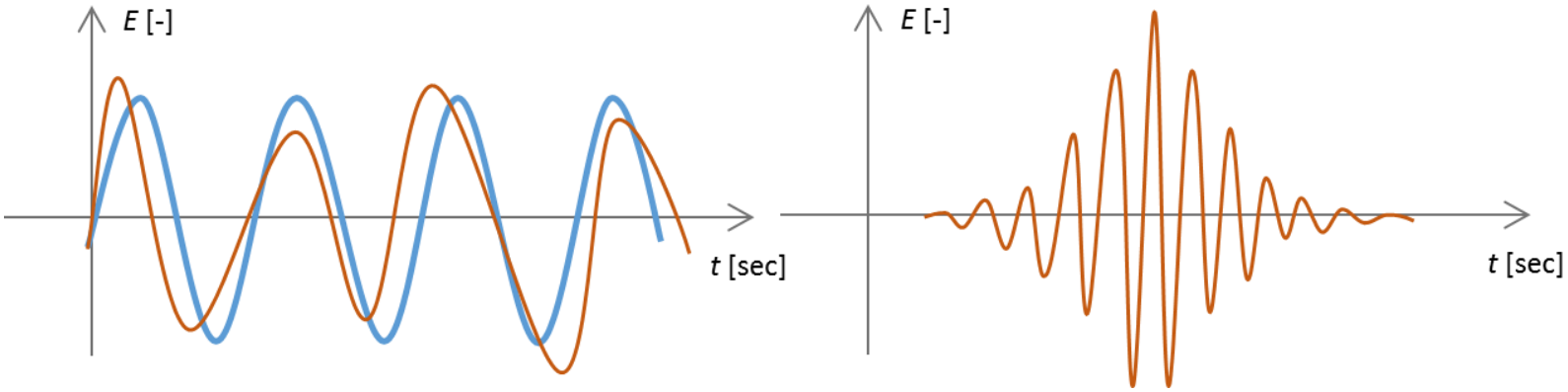
BMEatomfizika

OPTICS

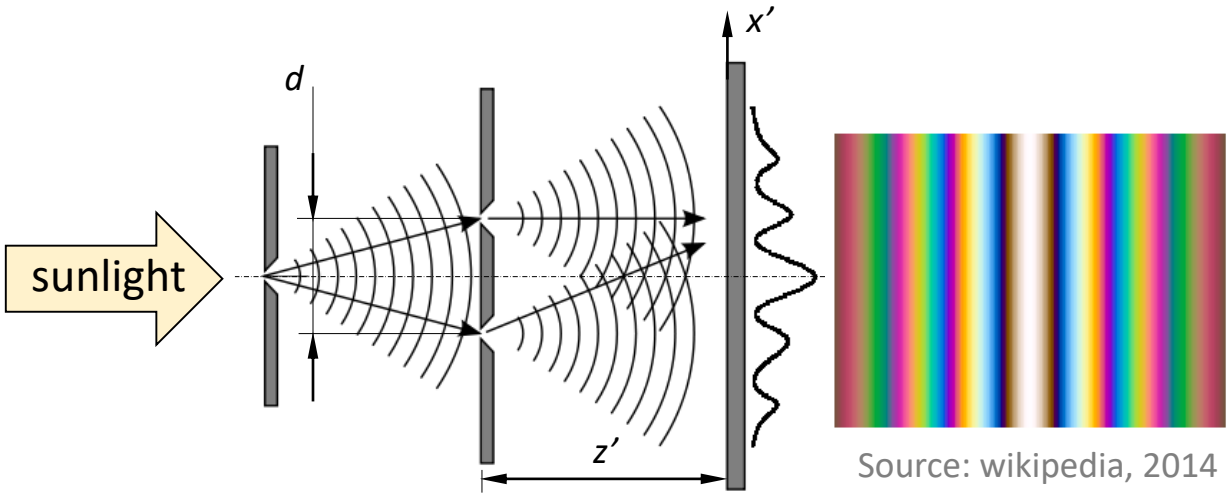
STATISTICAL OPTICS – TEMPORAL COHERENCE

Budapest University of Technology and Economics
Department of Atomic Physics, dr. Gábor Erdei

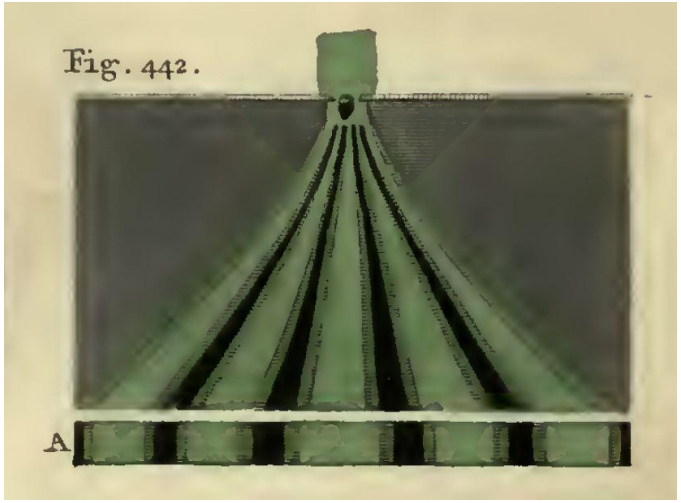
▪ Indeterministic oscillations ▪



▪ Young's two-slit interferometer ▪



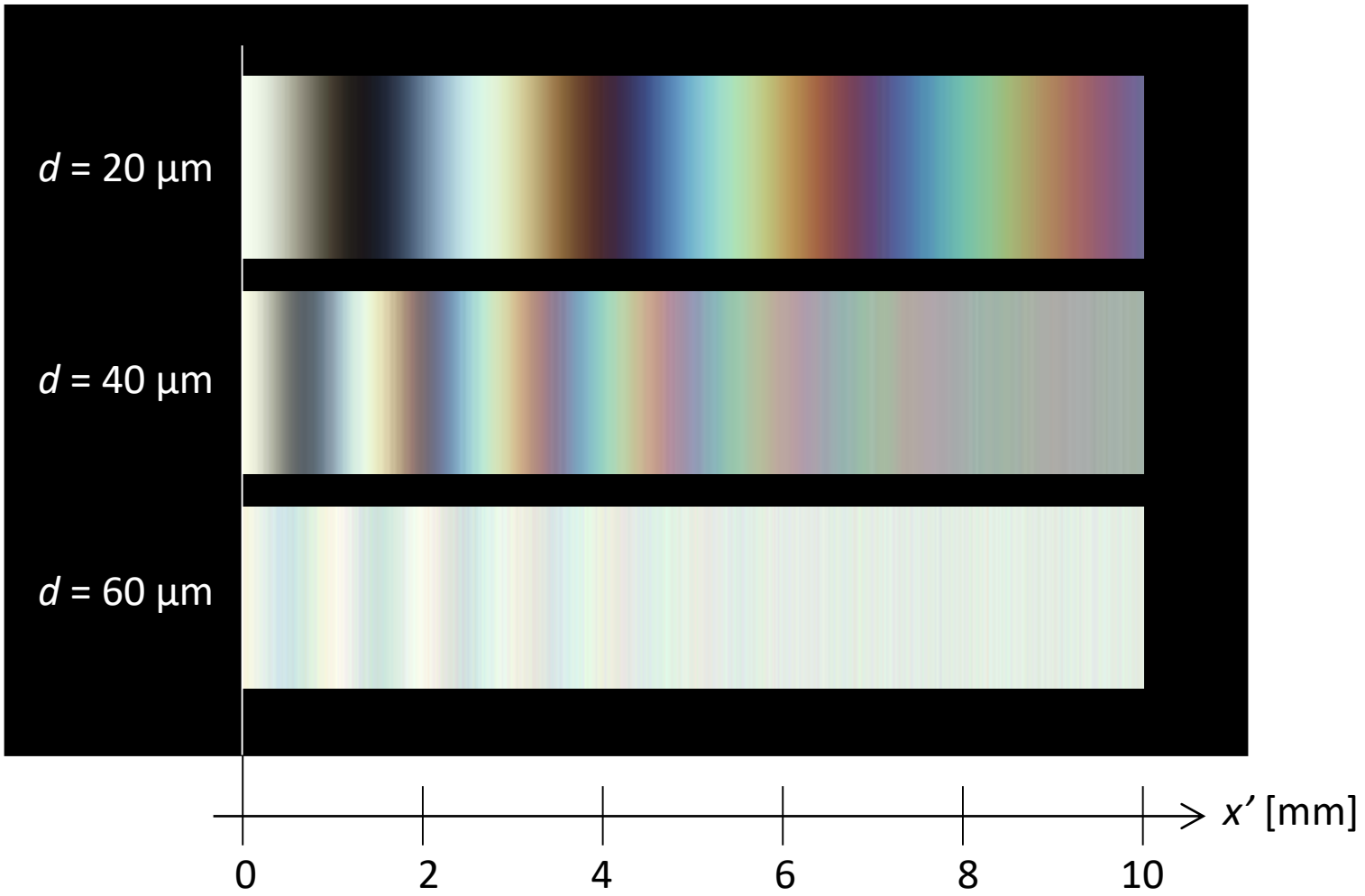
Source: wikipedia, 2014



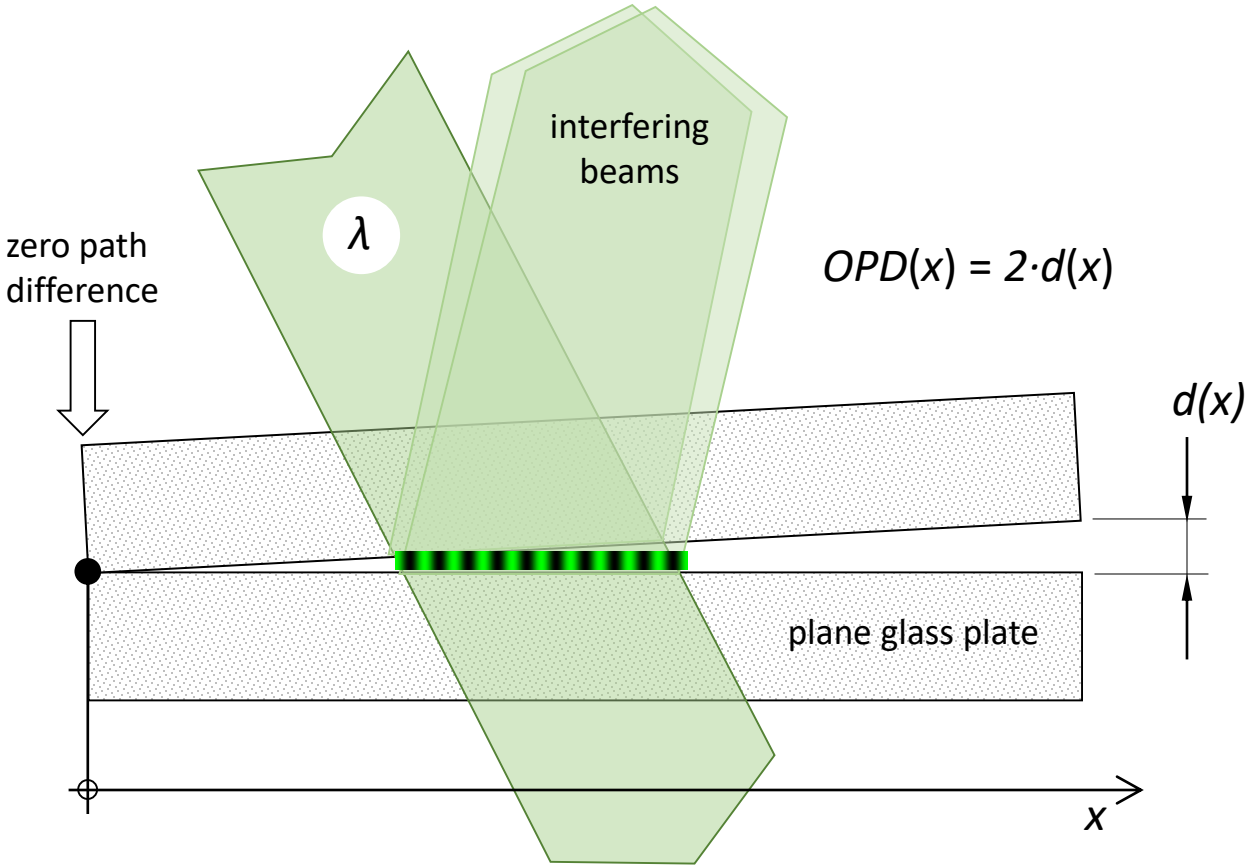
Source: T. Young, A Course of Lectures on Natural Philosphy and the Mechanical Arts, Vol. I, 1807

▪ Two-slit interference with unfiltered sunlight ▪

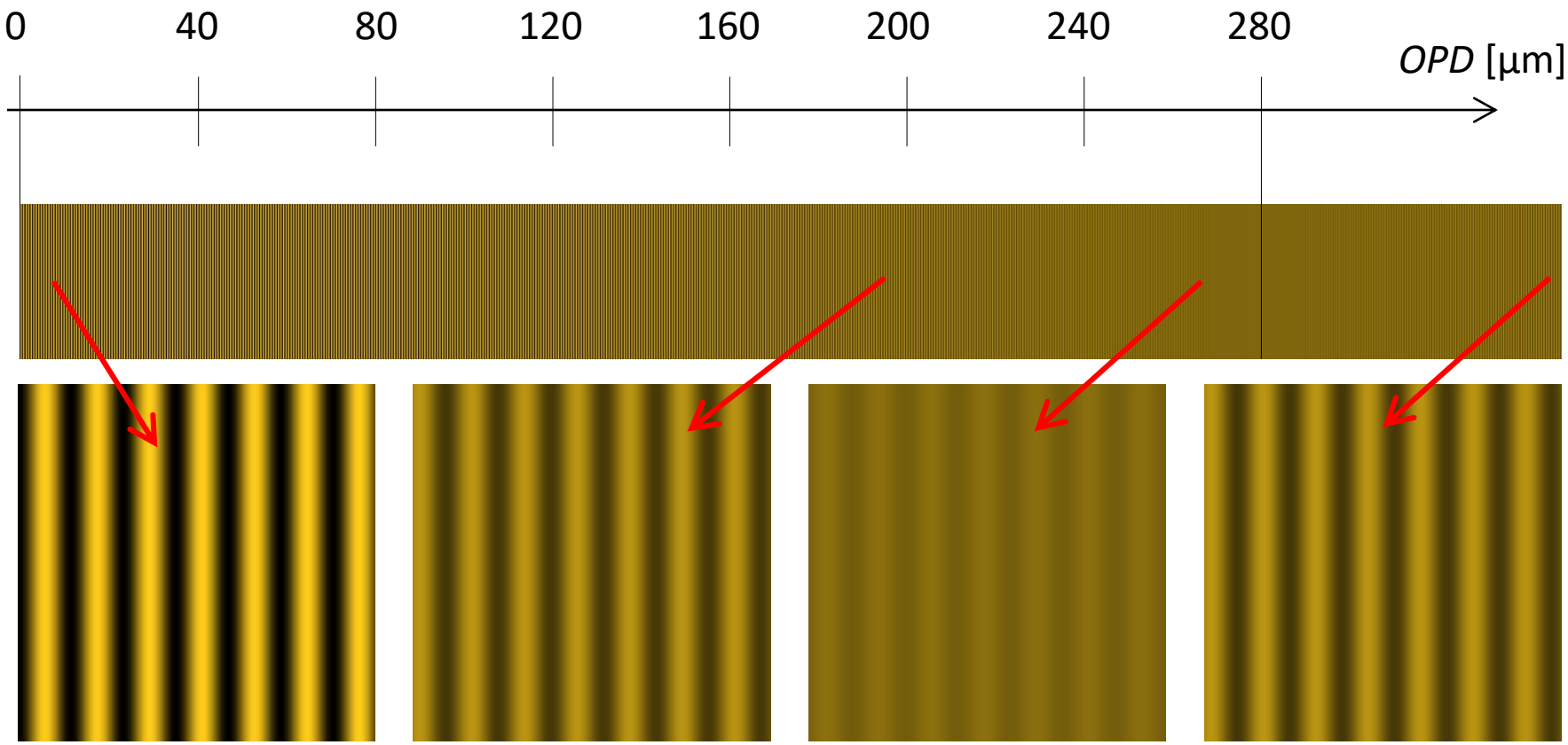
$z' = 100$ mm ; $\lambda = 400..700$ nm



▪ Fizeau interferometer – 1862 ▪



▪ Interference of sodium D-line (Fizeau) ▪



λ_1 = 589.0 nm
λ_2 = 589.6 nm

Visibility disappears
around fringe 492

▪ Two-beam interference at two wavelengths ▪

$$\left(\mathrm{E}_a(\boldsymbol{r},t)+\mathrm{E}_b(\boldsymbol{r},t)\right)^2 =$$
$$= A_1A_1\cos(\omega_1t+\Phi_{a1})\cdot\cos(\omega_1t+\Phi_{a1}) + A_1A_2\cos(\omega_1t+\Phi_{a1})\cdot\cos(\omega_2t+\Phi_{a2}) + A_1B_1\cos(\omega_1t+\Phi_{a1})\cdot\cos(\omega_1t+\Phi_{b1}) + A_1B_2\cos(\omega_1t+\Phi_{a1})\cdot\cos(\omega_2t+\Phi_{b2}) +$$
$$+ A_2A_1\cos(\omega_2t+\Phi_{a2})\cdot\cos(\omega_1t+\Phi_{a1}) + A_2A_2\cos(\omega_2t+\Phi_{a2})\cdot\cos(\omega_2t+\Phi_{a2}) + A_2B_1\cos(\omega_2t+\Phi_{a2})\cdot\cos(\omega_1t+\Phi_{b1}) + A_2B_2\cos(\omega_2t+\Phi_{a2})\cdot\cos(\omega_2t+\Phi_{b2}) +$$
$$+ B_1A_1\cos(\omega_1t+\Phi_{b1})\cdot\cos(\omega_1t+\Phi_{a1}) + B_1A_2\cos(\omega_1t+\Phi_{b1})\cdot\cos(\omega_2t+\Phi_{a2}) + B_1B_1\cos(\omega_1t+\Phi_{b1})\cdot\cos(\omega_1t+\Phi_{b1}) + B_1B_2\cos(\omega_1t+\Phi_{b1})\cdot\cos(\omega_2t+\Phi_{b2}) +$$
$$+ B_2A_1\cos(\omega_2t+\Phi_{b2})\cdot\cos(\omega_1t+\Phi_{a1}) + B_2A_2\cos(\omega_2t+\Phi_{b2})\cdot\cos(\omega_2t+\Phi_{a2}) + B_2B_1\cos(\omega_2t+\Phi_{b2})\cdot\cos(\omega_1t+\Phi_{b1}) + B_2B_2\cos(\omega_2t+\Phi_{b2})\cdot\cos(\omega_2t+\Phi_{b2})$$

$$\left\langle\left(\mathrm{E}_a(\boldsymbol{r},t)+\mathrm{E}_b(\boldsymbol{r},t)\right)^2\right\rangle =$$
$$= \frac{A_1A_1}{2} + \frac{A_1A_2}{2}\cos\left((\omega_1-\omega_2)t+\Phi_{a1}-\Phi_{a2}\right) + \frac{A_1B_1}{2}\cos\left((\omega_1-\omega_1)t+\Phi_{a1}-\Phi_{b1}\right) + \frac{A_1B_2}{2}\cos\left((\omega_1-\omega_2)t+\Phi_{a1}-\Phi_{b2}\right) +$$
$$+ \frac{A_2A_1}{2}\cos\left((\omega_2-\omega_1)t+\Phi_{a2}-\Phi_{a1}\right) + \frac{A_2A_2}{2} + \frac{A_2B_1}{2}\cos\left((\omega_2-\omega_1)t+\Phi_{a2}-\Phi_{b1}\right) + \frac{A_2B_2}{2}\cos\left((\omega_2-\omega_2)t+\Phi_{a2}-\Phi_{b2}\right) +$$
$$+ \frac{B_1A_1}{2}\cos\left((\omega_1-\omega_1)t+\Phi_{b1}-\Phi_{a1}\right) + \frac{B_1A_2}{2}\cos\left((\omega_1-\omega_2)t+\Phi_{b1}-\Phi_{a2}\right) + \frac{B_1B_1}{2} + \frac{B_1B_2}{2}\cos\left((\omega_1-\omega_2)t+\Phi_{b1}-\Phi_{b2}\right) +$$
$$+ \frac{B_2A_1}{2}\cos\left((\omega_2-\omega_1)t+\Phi_{b2}-\Phi_{a1}\right) + \frac{B_2A_2}{2}\cos\left((\omega_2-\omega_2)t+\Phi_{b2}-\Phi_{a2}\right) + \frac{B_2B_1}{2}\cos\left((\omega_2-\omega_1)t+\Phi_{b2}-\Phi_{b1}\right) + \frac{B_2B_2}{2}$$

▪ After averaging for a short time interval ▪

$$\left\langle \left(\mathrm{E}_a(\boldsymbol{r},t) + \mathrm{E}_b(\boldsymbol{r},t) \right)^2 \right\rangle =$$

$$= \frac{A_1 A_1}{2} + \frac{A_2 A_2}{2} + \frac{B_1 B_1}{2} + \frac{B_2 B_2}{2} +$$

$$+ \frac{A_1 B_1}{2}\cos\left(\Phi_{a1} - \Phi_{b1}\right) + \frac{A_2 B_2}{2}\cos\left(\Phi_{a2} - \Phi_{b2}\right) + \frac{B_1 A_1}{2}\cos\left(\Phi_{b1} - \Phi_{a1}\right) + \frac{B_2 A_2}{2}\cos\left(\Phi_{b2} - \Phi_{a2}\right) +$$

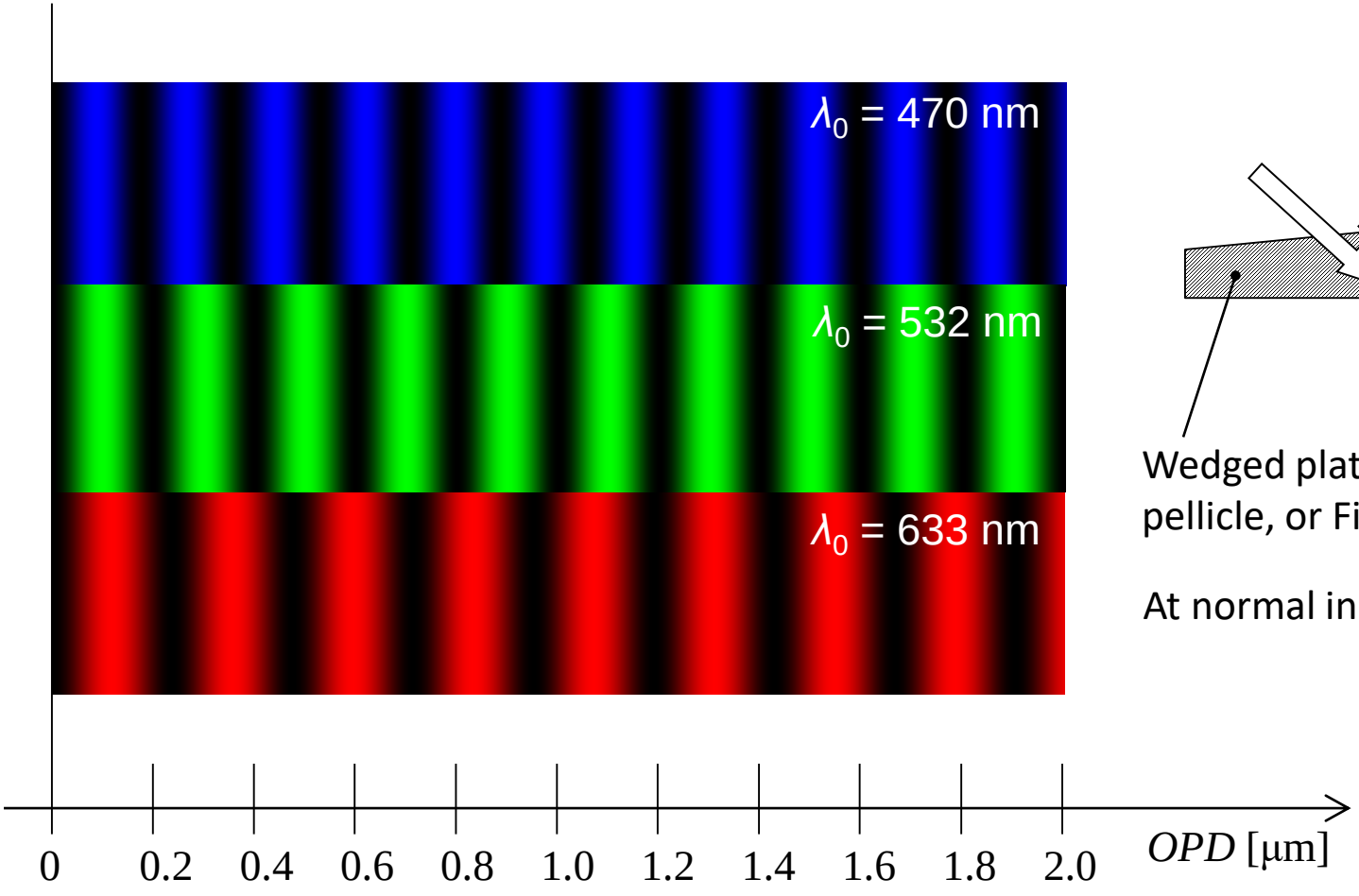
$$+ \frac{A_1 A_2}{2}\cos\left((\omega_1 - \omega_2)t + \Phi_{a1} - \Phi_{a2}\right) + \frac{A_1 B_2}{2}\cos\left((\omega_1 - \omega_2)t + \Phi_{a1} - \Phi_{b2}\right) +$$

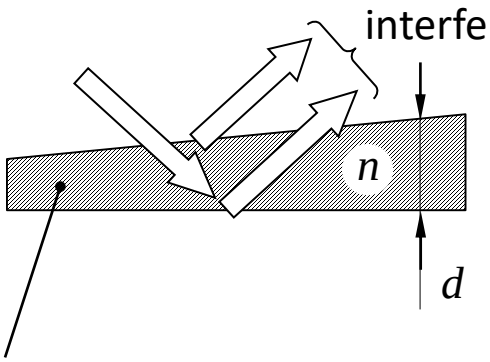
$$+ \frac{A_2 A_1}{2}\cos\left((\omega_2 - \omega_1)t + \Phi_{a2} - \Phi_{a1}\right) + \frac{A_2 B_1}{2}\cos\left((\omega_2 - \omega_1)t + \Phi_{a2} - \Phi_{b1}\right) +$$

$$+ \frac{B_1 A_2}{2}\cos\left((\omega_1 - \omega_2)t + \Phi_{b1} - \Phi_{a2}\right) + \frac{B_1 B_2}{2}\cos\left((\omega_1 - \omega_2)t + \Phi_{b1} - \Phi_{b2}\right) +$$

$$+ \frac{B_2 A_1}{2}\cos\left((\omega_2 - \omega_1)t + \Phi_{b2} - \Phi_{a1}\right) + \frac{B_2 B_1}{2}\cos\left((\omega_2 - \omega_1)t + \Phi_{b2} - \Phi_{b1}\right)$$

▪ Interference of discrete wavelengths ▪

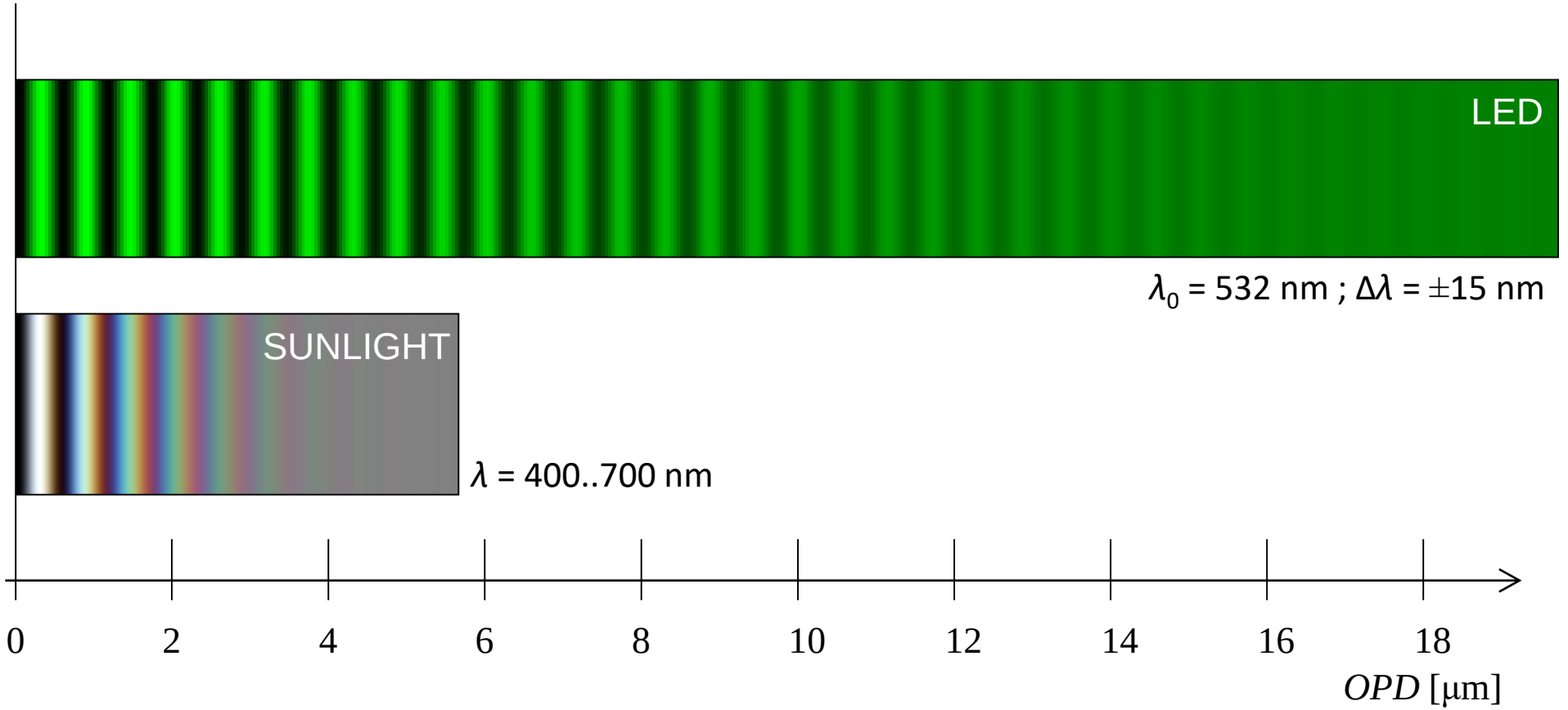




Wedged plate, e.g. soap bubble pellicle, or Fizeau interferometer

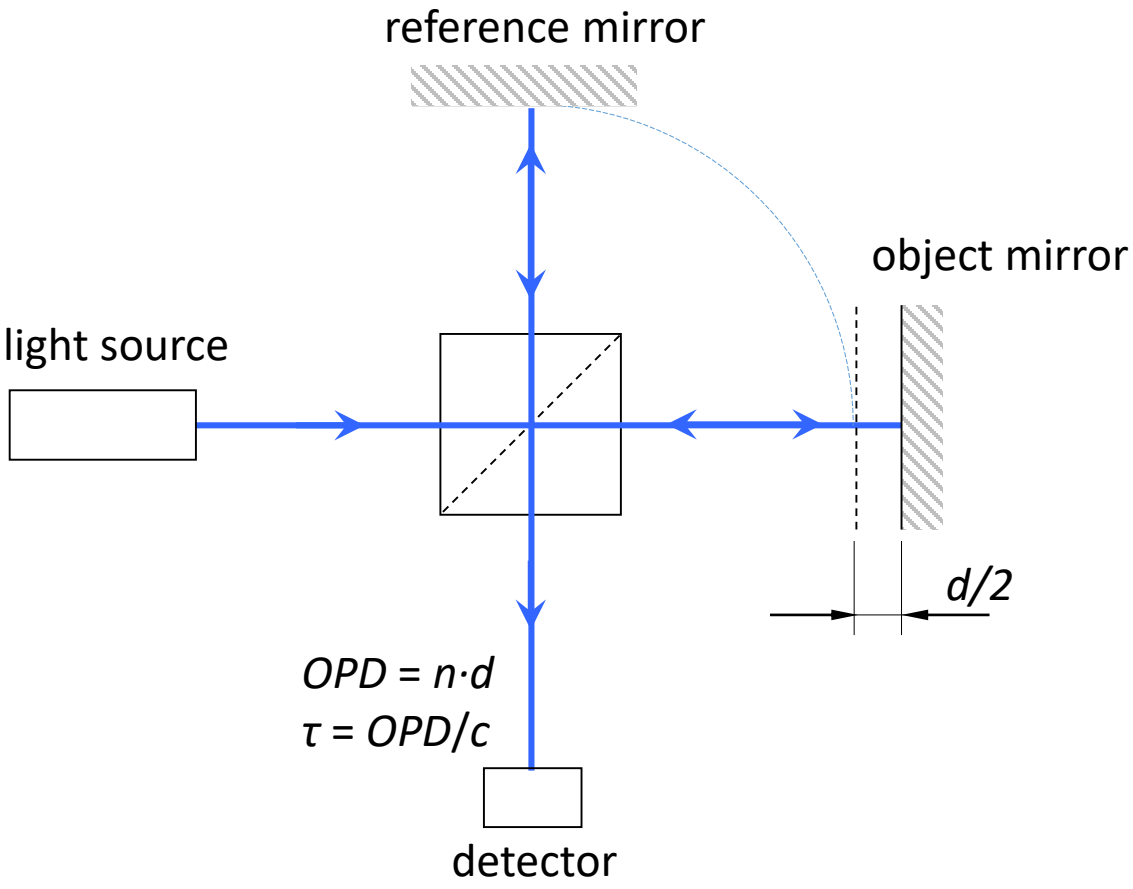
At normal incidence: $OPD = n \cdot 2d$

▪ Interference of waves with a continuous spectrum ▪

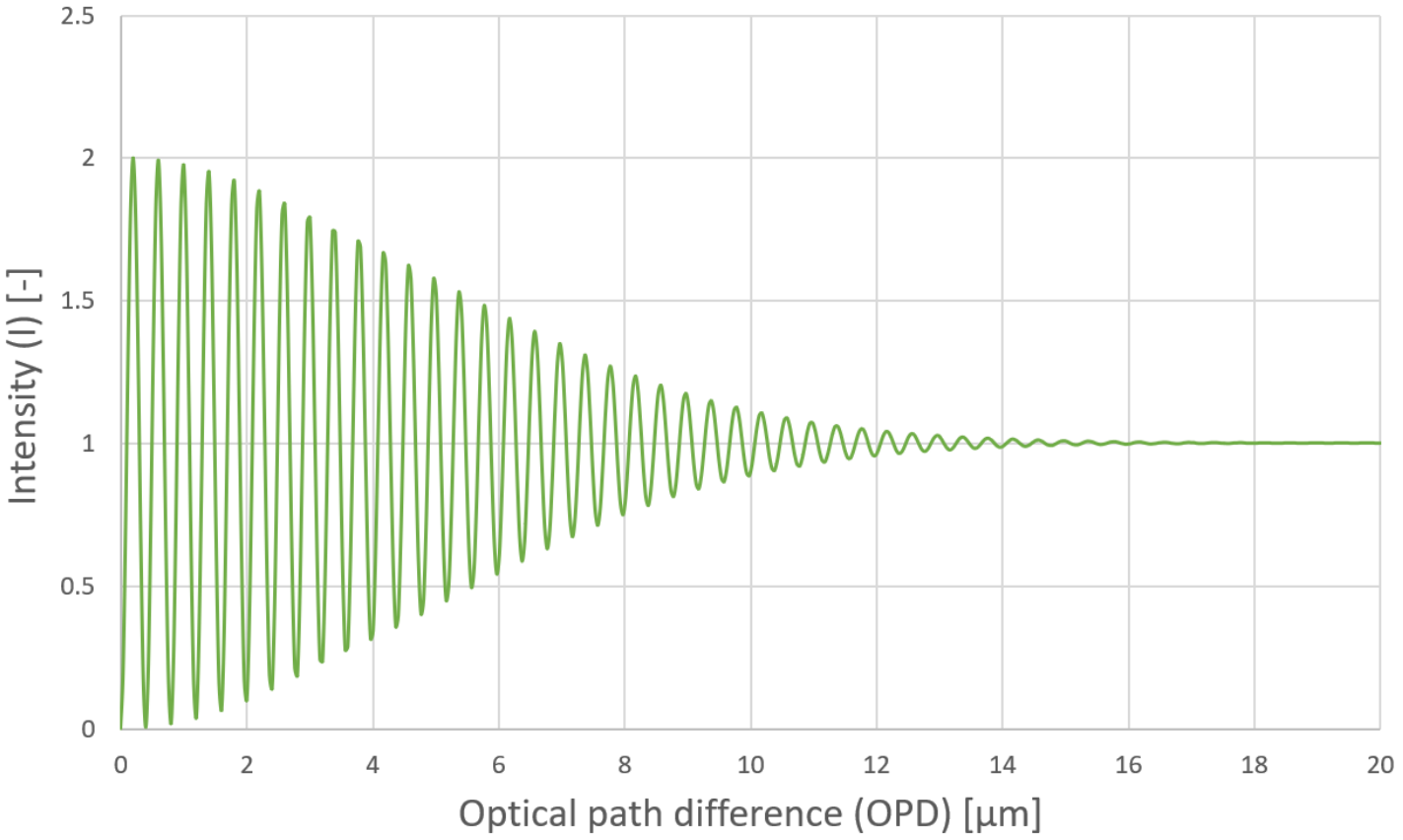


Where visibility disappears: $OPD = n \cdot l_L$

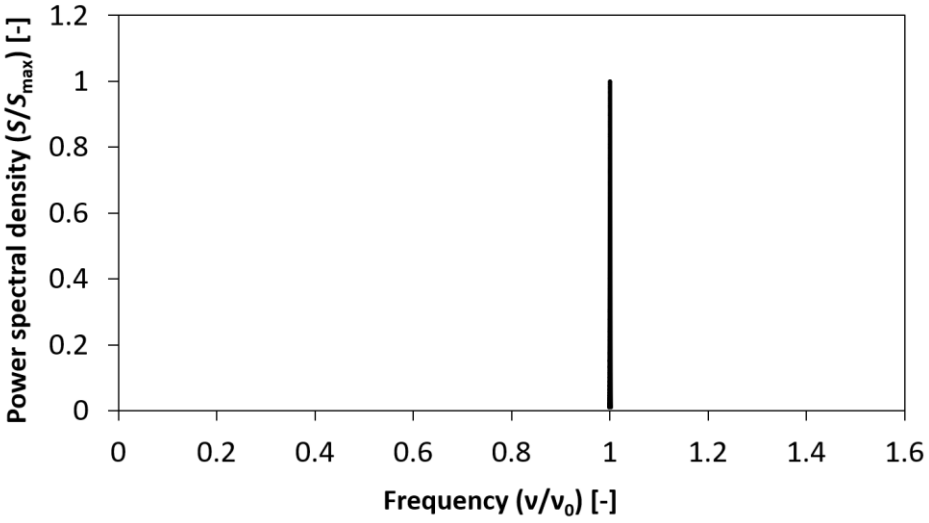
▪ Michelson (Twyman-Green) interferometer, 1880 ▪

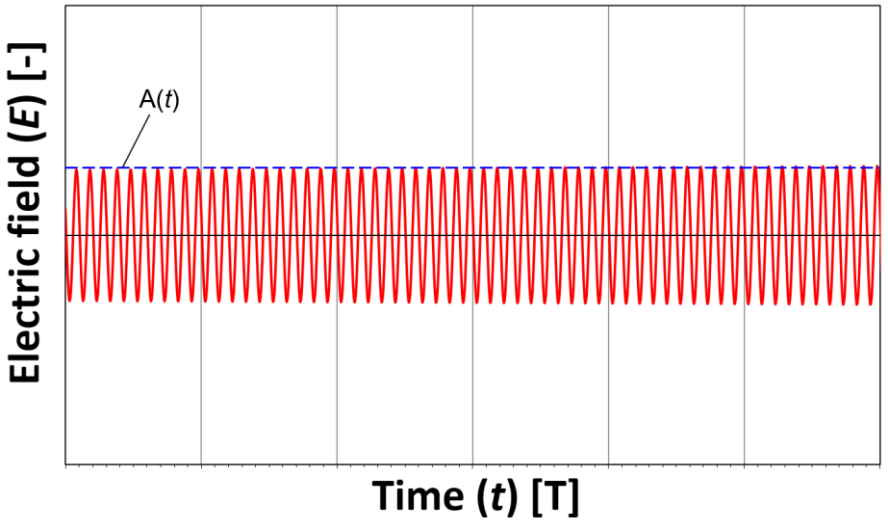


▪ Interferogram of a green LED ▪



▪ Deterministic (harmonic) oscillation ▪

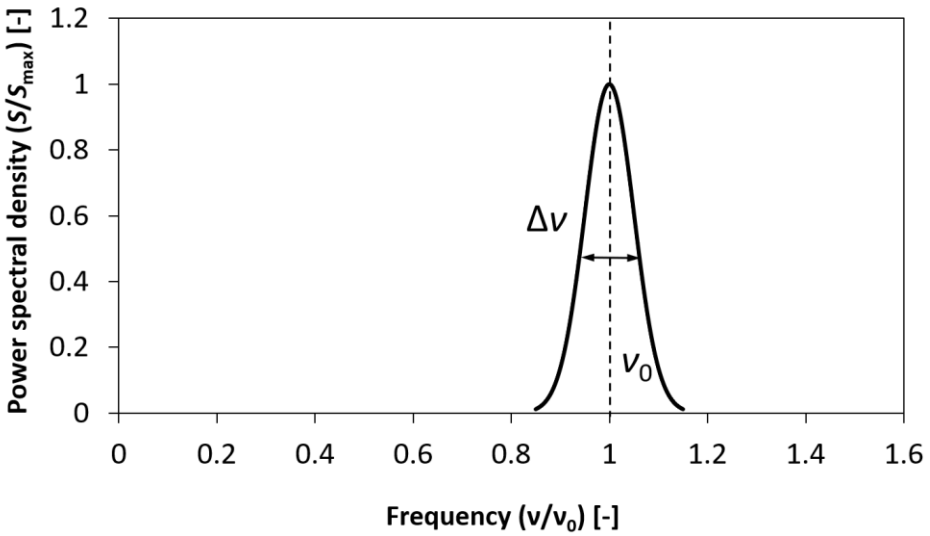




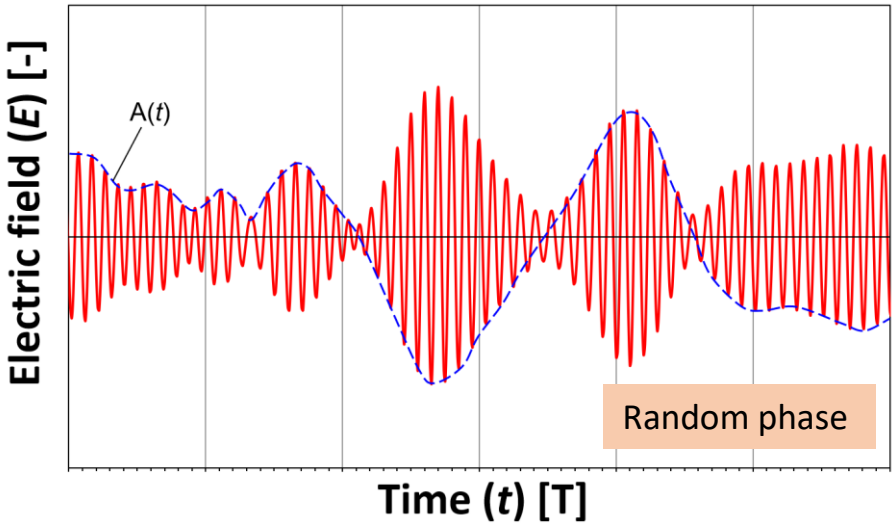
Partially coherent oscillation

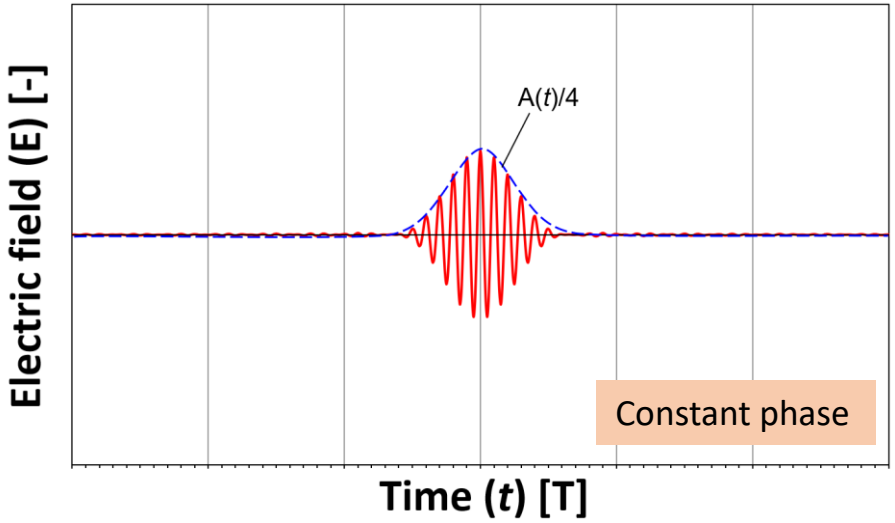
Quasi-monochromatic case $\Delta\nu << \nu_0$

Normal distribution



$\Delta\nu = 2\sigma = \nu_0/10$

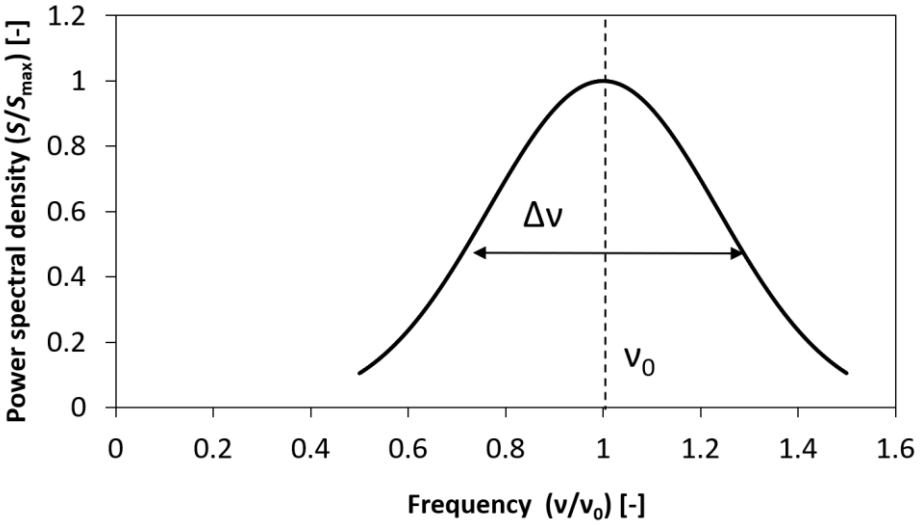




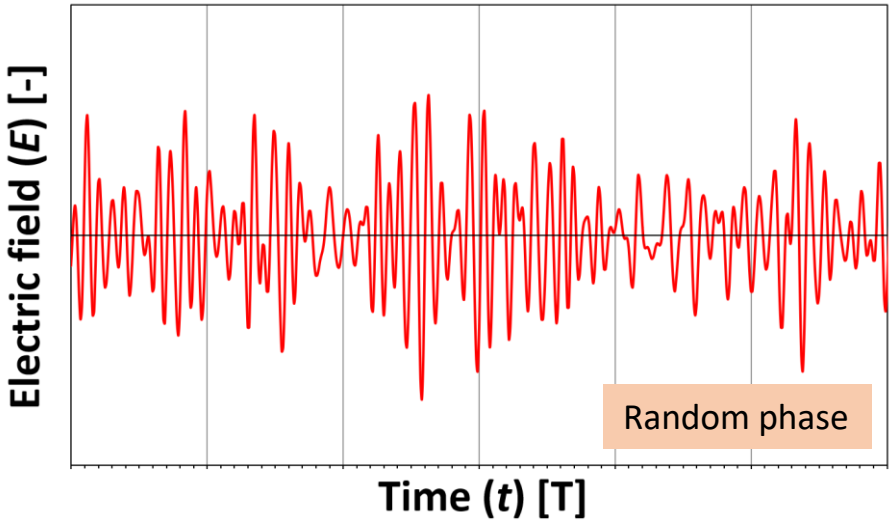
▪ Incoherent oscillation ▪

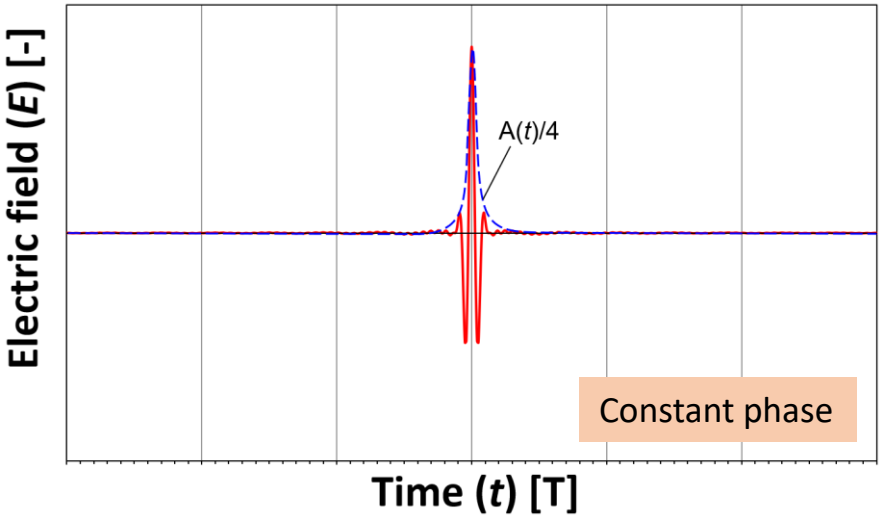
Incoherent case $\Delta\nu \approx> \nu_0$

Normal distribution



$\Delta\nu = 2\sigma = \nu_0/2$

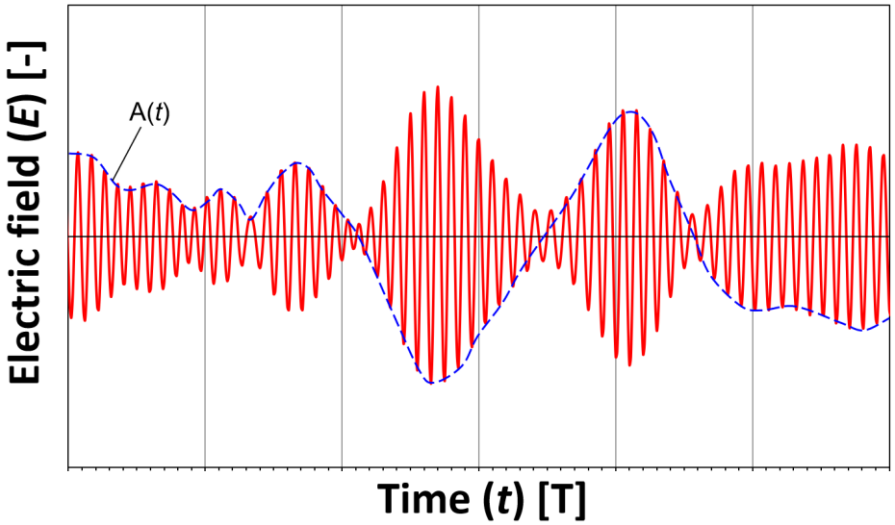




▪ Kinds of indeterministic signals ▪

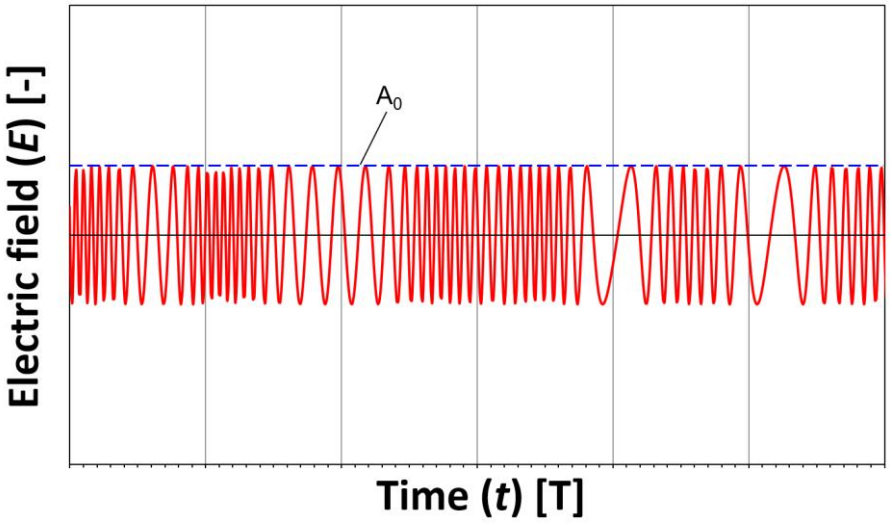
$$E(t) = A(t)\cdot\cos(\omega t + \varphi(t))$$

$E(t) = A(t)\cdot\cos(\omega t + \varphi_0)$



Typical visible radiation (thermal or chaotic light of spontaneous emission)

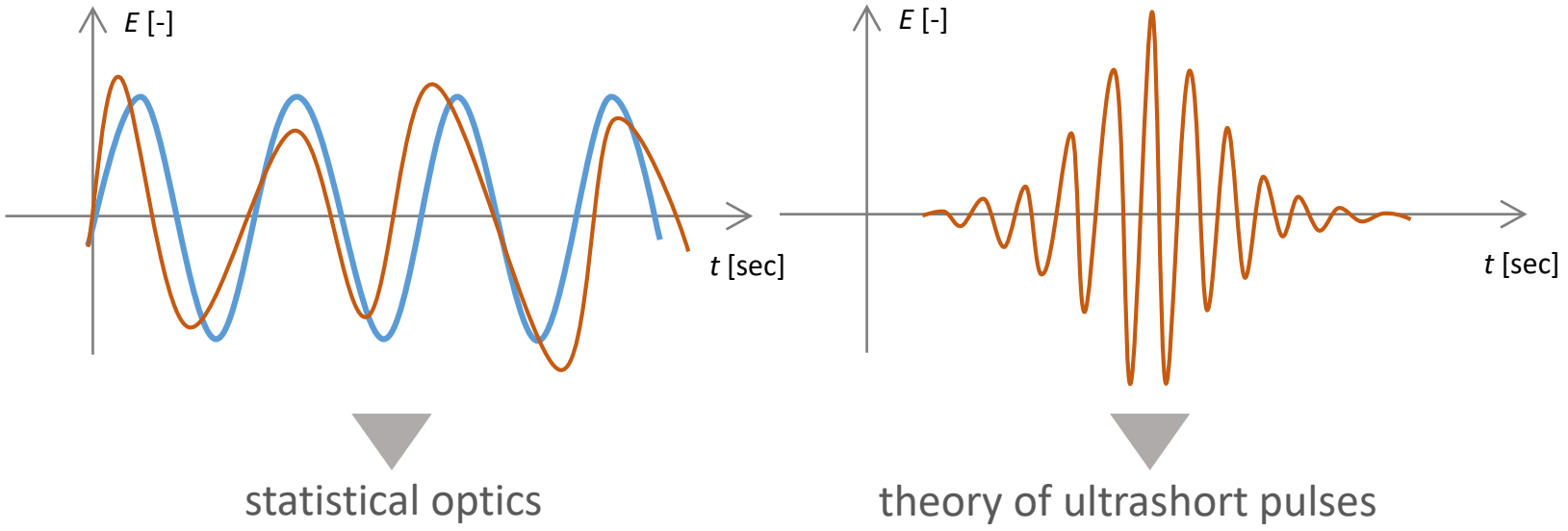
$E(t) = A_0\cdot\cos(\omega t + \varphi(t))$



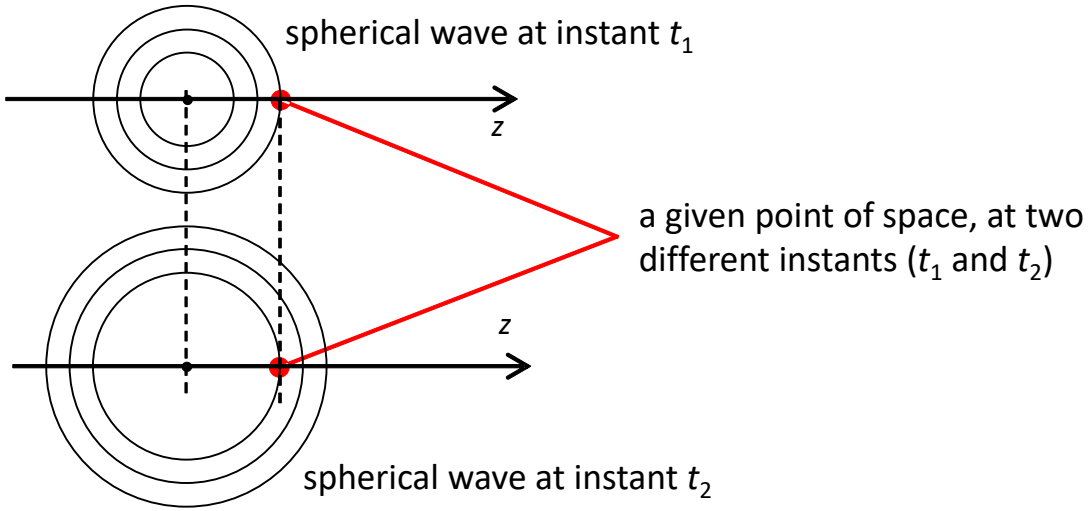
Single-mode lasers, FM-modulated radio waves (coherent light of induced emission)

▪ The concept of coherence (Laue, 1907) ▪

- Eigenfunctions of linear systems: harmonic (sinusoidal) functions
- The property of deviating from ideal periodicity: coherence
- Degrees of coherence: coherent, partially coherent and incoherent oscillation
- Causes of partial coherence: not perfectly harmonic wave, finite pulse
- Kinds of coherence: temporal and spatial coherence, polarization coherence
- Interpretation: examining the oscillation at two different spatial points or time instants how much consistency is in the phase states (how corellated they are)
- Temporal coherence: we examine the field behaviour at one point of space

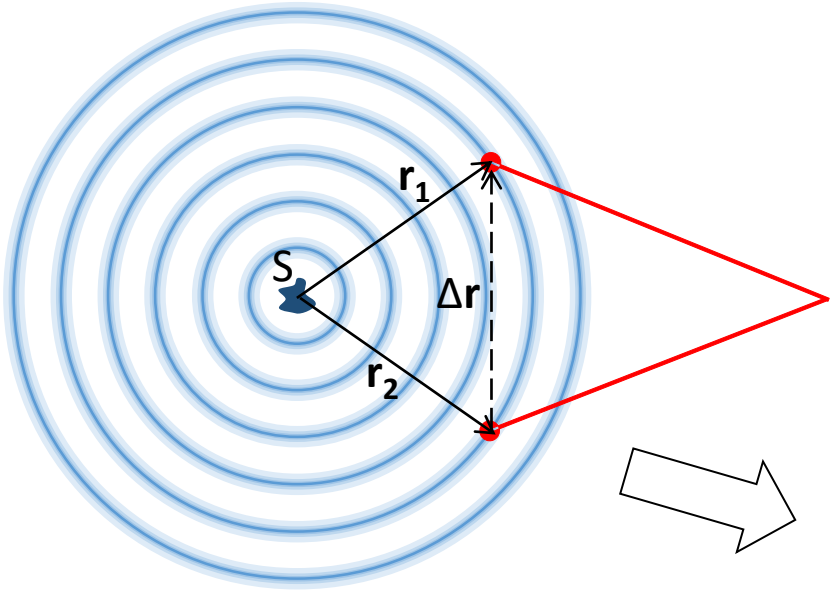


▪ Analysis of temporal coherence ▪



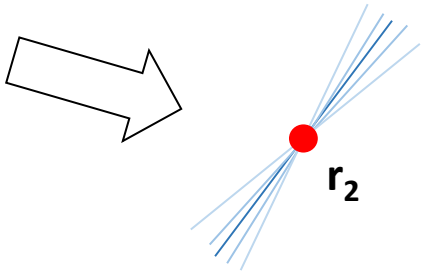
In order to analyze the oscillation at two different instants we need to split it into two components, and we need to delay one component by a $\tau = t_2 - t_1$ interval.

▪ Analysis of spatial coherence ▪



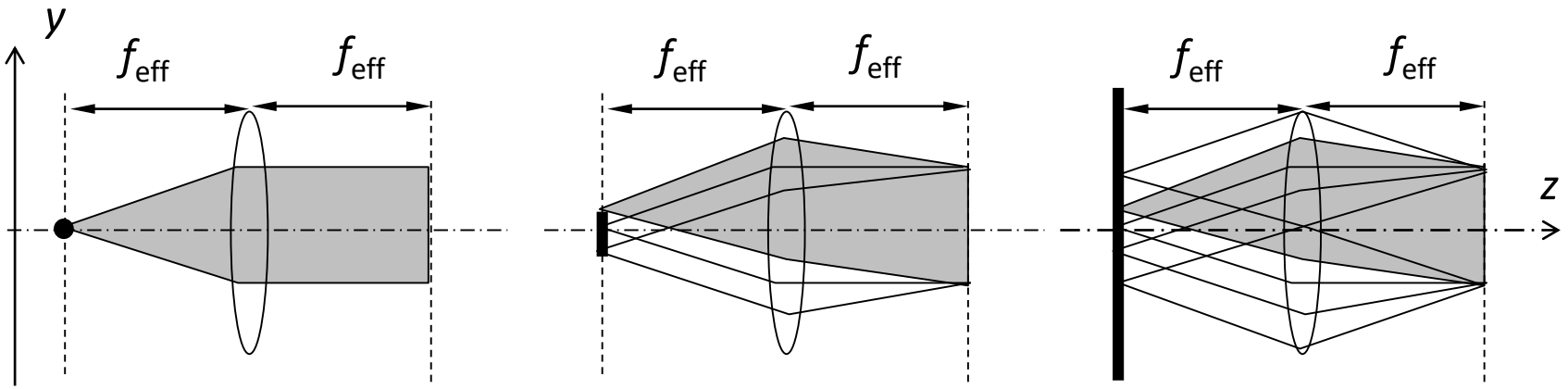
two arbitrary points of space ($\mathbf{r}_1$ and $\mathbf{r}_2$), at the same instant

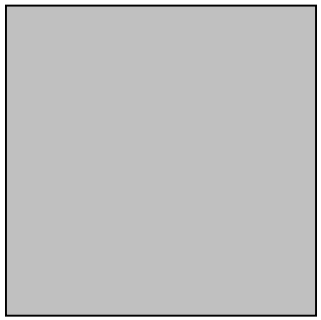
A wave started from an extended source at a given instant



Wavefronts originated from source "S" and passing through point $\mathbf{r}_2$

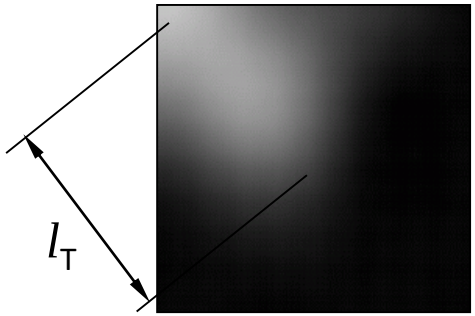
▪ Controlling spatial coherence ▪





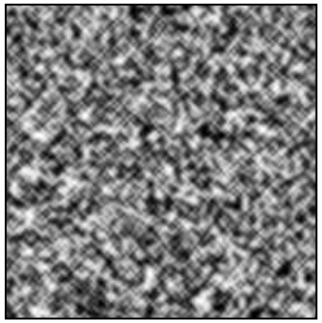
coherent case:
point source

$E_0 \approx$ const. ($l_T >> \lambda$)



partially coherent case:
small extended source

E_0 slowly changes($l_T > \lambda$)



incoherent case:
diffuse source

E_0 rapidly changes ($l_T \approx \lambda$)